Generating Functions Paper Summary

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In this paper, I will investigate the limiting distribution of the number of children of the root in a uniformly random rooted tree. I'll compare two classes of trees: plane trees, where the children of each vertex have a specified order, and Cayley trees, where the vertices are labelled, but the children are unordered. Their root degrees follow different probability distributions as the number of vertices tends to infinity.

I'll start with plane trees, whose OGF satisfies

$$T(z) = \frac{z}{1 - T(z)}.$$

Introducing a second variable u to keep track of the number of children of the root gives the bivariate generating function

$$T(z, u) = \frac{z}{1 - uT(z)}.$$

I'll show that, if X_n is the root degree of a uniformly random plane tree with n vertices, then for all fixed $k \geq 1$,

$$\lim_{n \to \infty} \mathbb{P}(X_n = k) = \frac{k}{2^{k+1}}.$$

Then, I'll consider Cayley trees, whose EGF is

$$T(z) = ze^{T(z)}.$$

Marking the root degree gives $T(z, u) = ze^{uT(z)}$. I will show that the root degree as the number of vertices tends to infinity is

$$\lim_{n \to \infty} \mathbb{P}(X_n = k) = \frac{e^{-1}}{(k-1)!}.$$

Finally, I'll conclude by showing how both distributions fit into Flajolet and Sedgewick's general framework of simply generated trees. For tree classes described by a generator of the form $T(z) = z\Phi(T(z))$ where Φ is a tree generator, they show that, under certain conditions, the limiting probability generating function for the root degree is

$$P(u) = \frac{u\Phi'(\tau u)}{\Phi'(\tau)},$$

where τ satisfies $\Phi(\tau) = \tau\Phi'(\tau)$. I'll show how this recovers the two distributions derived earlier.

The main reference is `https://algo.inria.fr/flajolet/Publications/book.pdf`.